

FieldSense PT-300 Pressure Transmitter -- Technical Manual

FIELD SENSE PT-300 TECHNICAL MANUAL

PRODUCT OVERVIEW

The FieldSense PT-300 is a state-of-the-art pressure transmitter designed specifically for water distribution networks and water treatment applications, including pump station discharge pressure monitoring and filter differential pressure measurement for backwash triggering. Engineered for precision and reliability, the PT-300 features advanced sensor technology that ensures accurate readings in various environmental conditions. Its robust construction and user-friendly interface make it an essential tool for operators in the water management sector.

TECHNICAL SPECIFICATIONS

- Measurement Range:
0 to 16 bar (0 to 232 psi)
- Accuracy:
±0.5% of full scale
- Output Signal:
4-20 mA (analog), HART protocol compatible
- Process Connection:
1/2" NPT male fitting or 1" Flange (DIN or ANSI standards)
- Ingress Protection:
IP68 rated for submersion up to 10 meters
- Hazardous Area Certification:
ATEX Zone 1 and 2, Class I, Div 1 and 2 for gas and dust environments

CALIBRATION PROCEDURE

To ensure optimal performance of the FieldSense PT-300 pressure transmitter, periodic calibration is recommended. The calibration procedure involves zero and span adjustments.

1. Zero Adjustment:
 - Connect the PT-300 to a pressure source that can be controlled.
 - Ensure the pressure is at zero (e.g., venting to atmosphere).
 - Access the transmitter's configuration menu via the HART communicator or the built-in display.
 - Select the ?Zero Adjustment? option and follow the prompts to confirm the zero reading.
 - Verify the display reads 0.00 bar (0.00 psi) after adjustment.
2. Span Adjustment:

- Apply a known reference pressure (e.g., 12 bar or 174 psi).
- Navigate to the ?Span Adjustment? option in the configuration menu.
- Input the known reference pressure value.
- The transmitter will calculate the necessary adjustment and prompt you to confirm.
- Check the output signal via the HART communicator to ensure it reflects the applied pressure accurately.

ERROR CODES

The FieldSense PT-300 is equipped with a diagnostic feature that displays error codes for troubleshooting. Below are common error codes, their causes, and recommended resolutions:

- Error Code E1: Sensor Failure

Cause: Internal sensor malfunction.

Resolution: Power cycle the transmitter. If the error persists, replace the unit.

- Error Code E2: Over Pressure

Cause: Pressure exceeds the maximum limit of 16 bar (232 psi).

Resolution: Reduce the process pressure. Inspect for potential pressure spikes.

- Error Code E3: Calibration Error

Cause: Calibration settings are out of expected range.

Resolution: Perform a calibration procedure as outlined above.

- Error Code E4: Communication Failure

Cause: Loss of signal between the transmitter and the control system.

Resolution: Check wiring connections and integrity of the HART communicator. Ensure proper configuration settings are in place.

- Error Code E5: Temperature Out of Range

Cause: Ambient temperature exceeds the operating limits.

Resolution: Ensure the operating environment is within -40°C to +85°C (-40°F to +185°F). Relocate the transmitter if necessary.

MAINTENANCE

Routine maintenance of the FieldSense PT-300 is essential for ensuring long-term reliability and accuracy. Follow these guidelines for effective maintenance:

1. Routine Checks:

- Inspect the physical condition of the transmitter and process connections for signs of wear, corrosion, or leaks.
- Verify that the display and output signals are functioning correctly by performing a quick pressure test on the system.
- Clean the transmitter housing with a damp cloth to remove dust and contaminants.

2. Troubleshooting:

- If the transmitter exhibits abnormal readings, perform a zero and span check to verify calibration.
- In the case of persistent error codes, consult the error code section above for specific resolutions.

- If operational issues continue, contact FieldSense technical support for further assistance.

3. Replacement Parts:

- Periodically inspect and replace seals and gaskets to maintain optimal performance and prevent leaks.
- Use only genuine FieldSense replacement parts to ensure compatibility and reliability.

By following this technical manual, operators can maintain the FieldSense PT-300 in optimal working condition, ensuring accurate pressure monitoring in water distribution and treatment applications.